

Position Paper:

Where AI assistance adds value

There are actually tons of scenarios where we can fit the AI assistance according to me. More or less it can be fit anywhere in our daily life, but it is absolutely not a wise decision to fit AI everywhere without thinking of the constraints and boundaries that actually AI have. So when does AI assistance add value. According to my opinion, it adds, either it is solving a problem more efficiently and cheaply, both in terms of financially and timely. I mean the AI should only be given autonomy when the cost of the running LLMs tokens are actually worth the benefits that it is providing and that could be in any form. So for example, it takes a hacker 60s to break into the system, but it takes the AI only 5s to break into the system, and those 5 seconds costs 20 dollars. Then it is worth it, because it is saving both the time and reducing risk of letting know. But if AI is taking 20 dollars just to investigate the tiny issue that is not urgent then it is not worth it. If I specifically talk about the areas where the AI can add huge value, I would say, in the cyber security field, it can add a lot of value, because many attacks happen at machine speed, making it impossible for humans to react quickly enough. One of the biggest advantages of AI is its ability to process and analyze massive amounts of data within seconds. For example, AI can continuously monitor network traffic, identify unusual login attempts, detect malware signatures, and filter thousands of security alerts in real time. Instead of replacing security analysts, AI reduces their workload by eliminating false alarms and highlighting only the most suspicious activities that require attention. Another area where AI adds significant value is threat intelligence. AI can collect information from multiple sources, compare attack patterns with previous incidents, and identify indicators of compromise much faster than a human analyst. It can also generate summaries of security logs, recommend possible mitigation steps, and prioritize incidents based on their potential impact. By automating these repetitive and time-consuming tasks, AI improves the efficiency of security operations while allowing human experts to concentrate on strategic analysis and decision-making.

when AI must not decide

Although AI can greatly improve the efficiency of security operations, there are situations where it must never make the final decision on its own. In my opinion, AI should not decide whenever a decision is irreversible, carries significant legal or ethical consequences, or has the potential to cause widespread damage if it is wrong. Unlike humans, AI makes decisions based on patterns in data rather than true understanding, common sense, or moral reasoning. It cannot fully understand human values, unusual situations, or the broader context behind an incident. For example, an AI system may detect suspicious activity on a company's network and recommend blocking a user account. However, permanently deleting critical data, shutting down an organization's entire infrastructure, exposing confidential information, or launching a cyber counterattack should never be left entirely to AI. A false prediction or an unexpected situation could cause financial losses, disrupt essential services, or even affect people's safety. Once these actions are taken, many of their consequences cannot be easily reversed. Similarly, AI should not make decisions in situations where accountability is required. If an autonomous security decision causes harm, the AI cannot accept legal responsibility or explain its ethical reasoning. Only human decision-makers can be held accountable for the outcomes of such actions. Therefore,

while AI can provide valuable recommendations and risk assessments, the final authority for high-impact security decisions should always remain with qualified human professionals.

when Human Approval is mandatory

Human approval should be mandatory whenever a security action has the potential to significantly impact an organization's operations, finances, reputation, or the privacy and safety of individuals. While AI is highly effective at detecting threats, analyzing large volumes of data, and recommending appropriate responses, the final approval should remain with trained security professionals for high-impact decisions. This ensures that important actions are evaluated not only from a technical perspective but also from legal, ethical, and business viewpoints. For example, actions such as shutting down critical servers, blocking access to essential business systems, deleting important data, sharing sensitive information with external authorities, or responding to a large-scale cyberattack should always require human approval before execution. A security analyst can review the AI's recommendation, verify the available evidence, assess the potential consequences, and determine whether the proposed action is appropriate for the specific situation. This extra layer of review helps prevent costly mistakes caused by false positives or unexpected circumstances. In my opinion, the most effective security strategy is one where AI acts as an intelligent assistant rather than an independent decision-maker. AI should continuously monitor systems, identify suspicious activities, and provide recommendations in real time, while humans remain responsible for approving critical actions that could have lasting consequences. This human-in-the-loop approach combines the speed and analytical capabilities of AI with human judgment, accountability, and experience, resulting in more reliable and trustworthy security operations.

Conclusion

AI should be viewed as a powerful assistant rather than a complete replacement for human decision-makers in security operations. It provides the greatest value when performing repetitive, time-sensitive, and data-intensive tasks such as monitoring systems, detecting threats, and analyzing large amounts of security data. However, decisions that are irreversible, legally sensitive, ethically complex, or capable of causing significant organizational damage should never be made by AI alone. In such situations, human approval is essential to ensure accountability, critical thinking, and sound judgment. Therefore, the most effective approach is a human-in-the-loop model, where AI enhances the speed and efficiency of security operations while humans retain responsibility for all critical decisions. This balanced division of responsibility allows organizations to benefit from AI's capabilities without compromising safety, ethics, or accountability.

AI assistance note:

I have used Gemini for brainstorming the ideas and to get the basic understanding of the things. I have also rewritten some content to the AI too to better phrase the sentences.